RATIO AND ROOT TESTS

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Name: Solutions

Determine whether the following series converge or diverge.

1. $\sum_{n=1}^{\infty} \frac{n}{5^n}$

$$\lim_{n\to\infty} \left| \frac{(n+1)}{5^{(n+1)}} \frac{5^n}{n} \right| = \lim_{n\to\infty} \frac{n+1}{5n} = \lim_{n\to\infty} \left(\frac{1}{5} + \frac{1}{5n}\right) = \frac{1}{5} + 0 < 1$$

$\Rightarrow \sum_{n=1}^{\infty} \frac{n}{5^n}$ converges by the Ratio Test.

2. $\sum_{k=1}^{\infty} \frac{1}{k!}$

$$\lim_{k\to\infty} \left| \frac{1}{(k+1)!} \frac{k!}{1} \right| = \lim_{k\to\infty} \frac{k!}{(k+1)k!} = \lim_{k\to\infty} \frac{1}{k+1} = 0$$

$\Rightarrow \sum_{k=1}^{\infty} \frac{1}{k!}$ converges by the Ratio Test.

3. $\displaystyle\sum_{n=1}^{\infty} \frac{10^n}{(n+1)4^{2n+1}}$

$2(n+1)+1 = 2n+2+1$
$= 2n+3$

$$\lim_{n\to\infty} \left| \frac{10^{n+1}}{(n+2)4^{2n+3}} \frac{(n+1)4^{2n+1}}{10^n} \right| = \lim_{n\to\infty} \frac{10^{n+1}}{10^n} \frac{4^{2n+1}}{4^{2n+3}} \frac{n+1}{n+2} = \lim_{n\to\infty} 10\left(\frac{1}{4^2}\right)\frac{n+1}{n+2} = \frac{10}{16}(1) < 1$$

$\Rightarrow \displaystyle\sum_{n=1}^{\infty} \frac{10^n}{(n+1)4^{2n+1}}$ converges by the Ratio Test.

4. $\displaystyle\sum_{n=1}^{\infty} \left(\frac{n^2+1}{2n^2+1}\right)^n$

$$\lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{n^2+1}{2n^2+1}\right)\right|^n} = \lim_{n\to\infty} \frac{n^2+1}{2n^2+1} = \frac{1}{2} < 1$$

$\Rightarrow \displaystyle\sum_{n=1}^{\infty} \left(\frac{n^2+1}{2n^2+1}\right)^n$ Converges by the Root Test.

5. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(\ln(n))^n}$

$$\lim_{n\to\infty} \sqrt[n]{\left|\frac{(-1)^{n-1}}{\ln(n)^n}\right|} = \lim_{n\to\infty} \frac{1}{\sqrt[n]{\ln(n)^n}} = \lim_{n\to\infty} \frac{1}{\ln(n)} = 0 < 1$$

$\Rightarrow \displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\ln(n)^n}$ converges absolutely by the Root Test.

6. $\displaystyle\sum_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^{n^2}$

$$\lim_{n\to\infty} \sqrt[n]{\left(1+\frac{1}{n}\right)^{n^2}} = \lim_{n\to\infty} \left(1+\tfrac{1}{n}\right)^{n^2/n} = \lim_{n\to\infty} \left(1+\frac{1}{n}\right)^{n} = e > 1$$

$\Rightarrow \displaystyle\sum_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^{n^2}$ diverges by the Root Test.